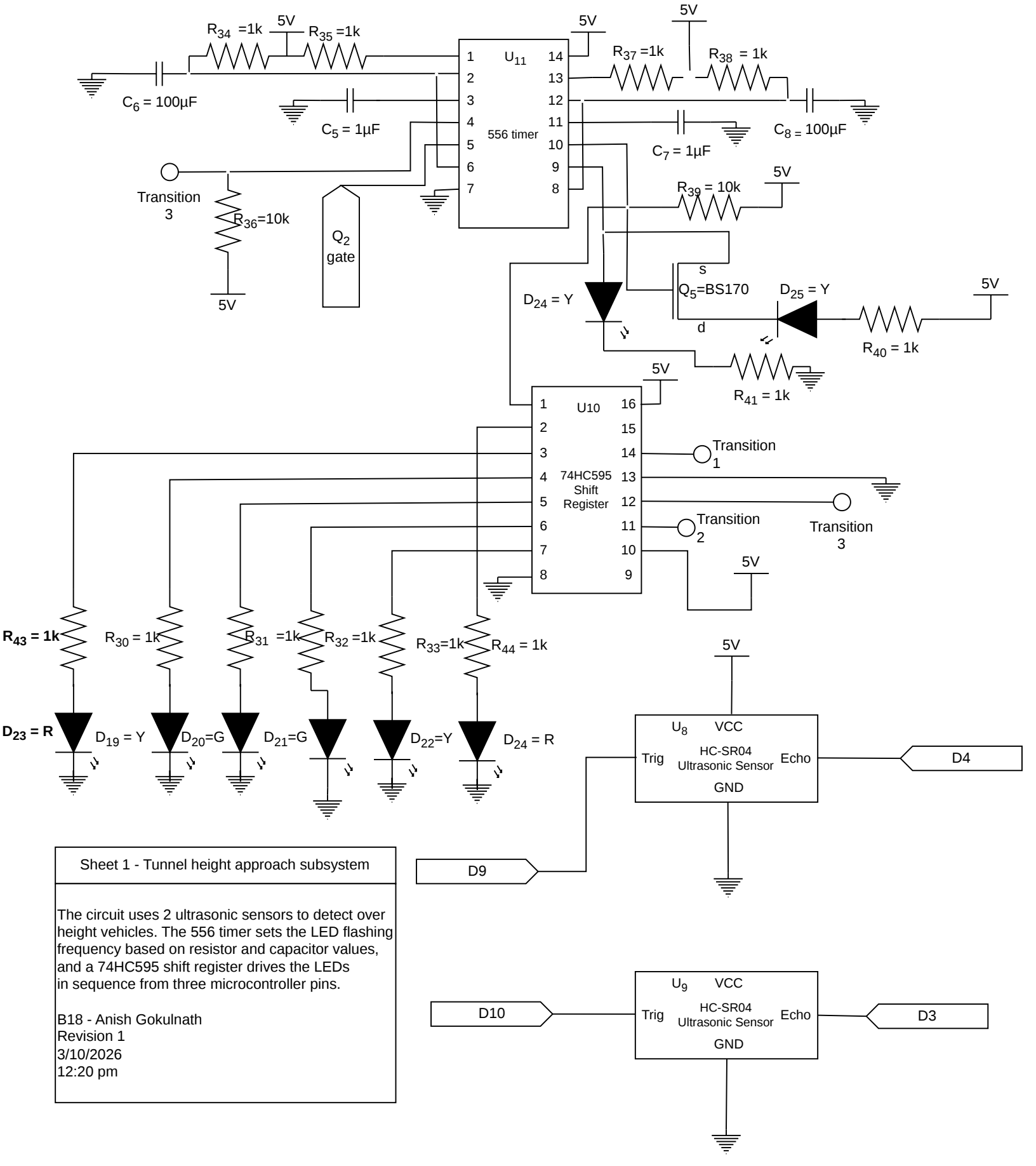


Sheet 1 - Tunnel height approach subsystem

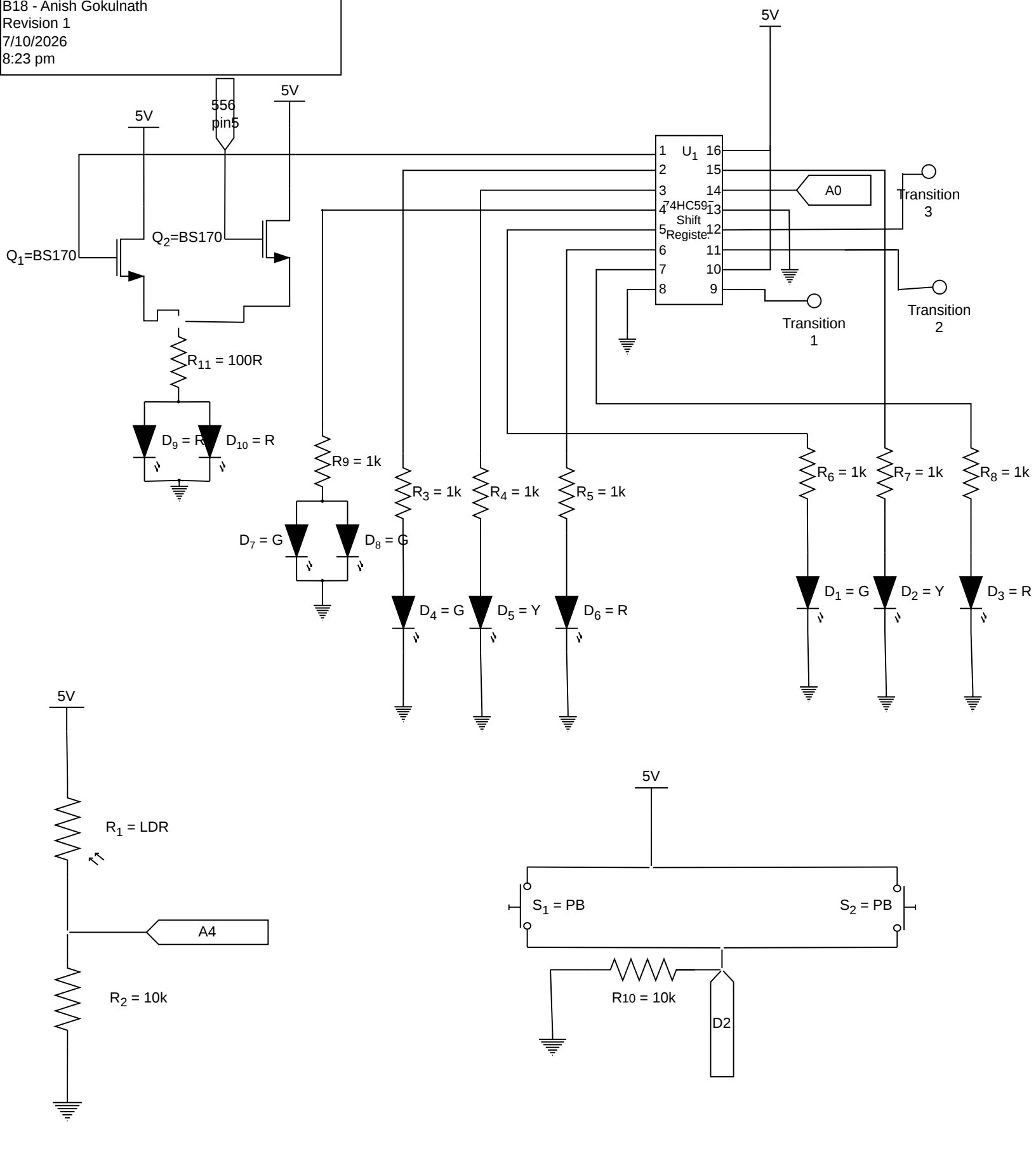
The circuit uses 2 ultrasonic sensors to detect over height vehicles. The 556 timer sets the LED flashing frequency based on resistor and capacitor values, and a 74HC595 shift register drives the LEDs in sequence from three microcontroller pins.

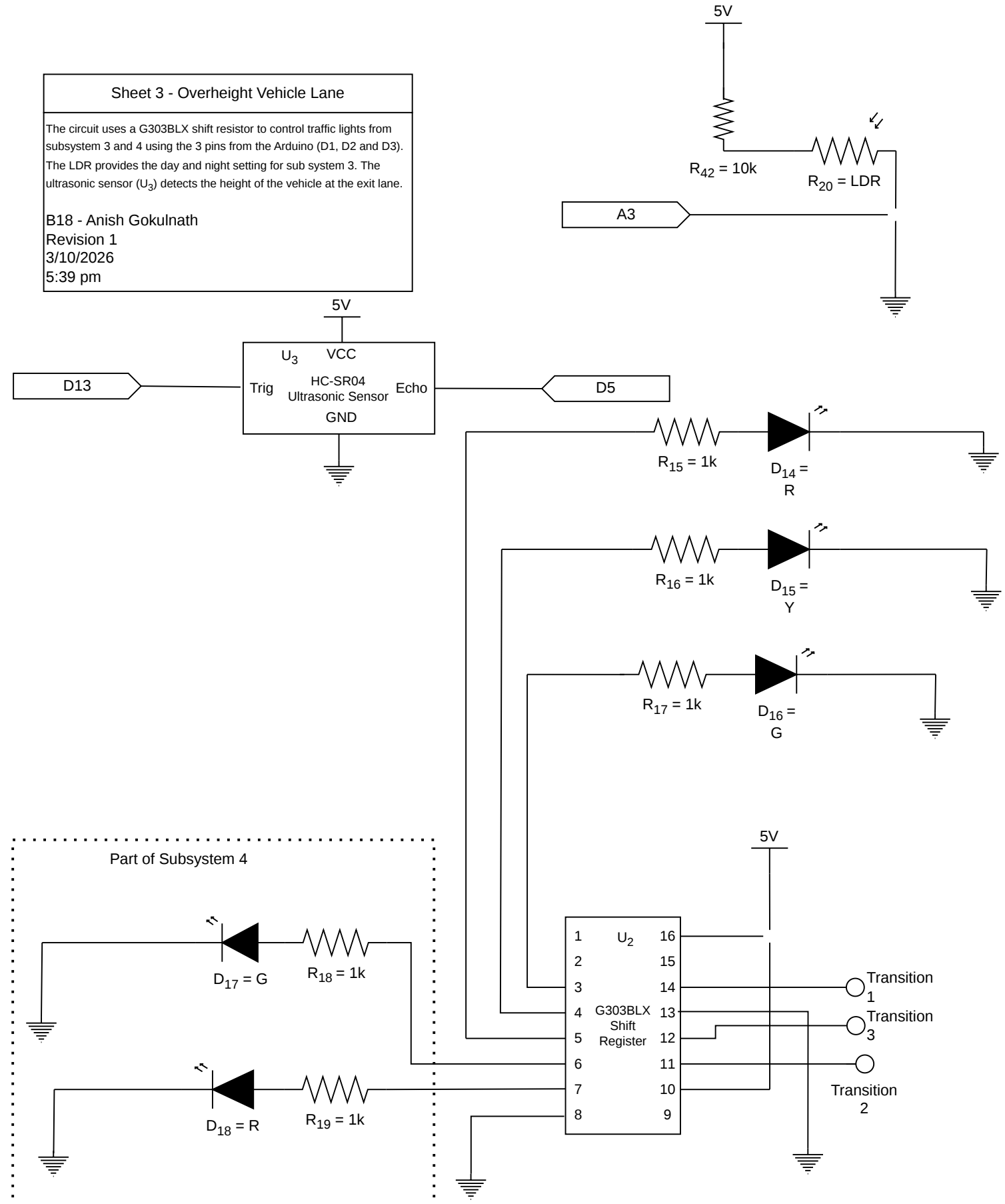
B18 - Anish Gokulnath
Revision 1
3/10/2026
12:20 pm

Sheet 2 - Tunnel Avenue control subsystem

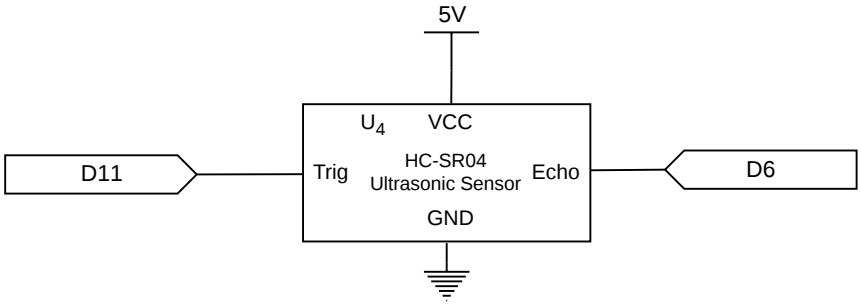
This circuit uses a 74HC595 shift register to drive two traffic lights, with an LDR for day/night sensing and push buttons as inputs.

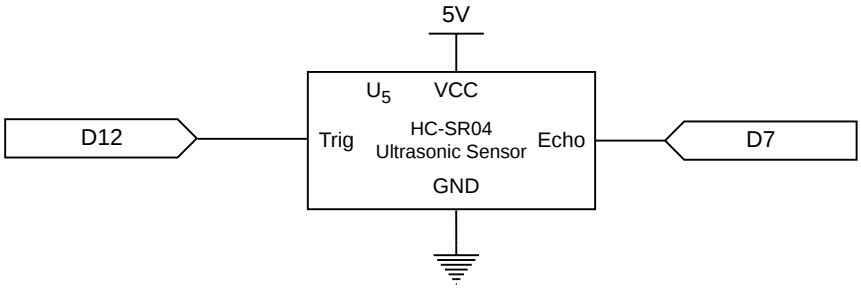
B18 - Anish Gokulnath
Revision 1
7/10/2026
8:23 pm




Sheet 3 - Overheight Vehicle Lane
The circuit uses a G303BLX shift resistor to control traffic lights from subsystem 3 and 4 using the 3 pins from the Arduino (D1, D2 and D3). The LDR provides the day and night setting for sub system 3. The ultrasonic sensor (U_3) detects the height of the vehicle at the exit lane.
B18 - Anish Gokulnath
Revision 1
3/10/2026
5:39 pm
5V
R_{42} = 10k
R_{20} = LDR
A3
5V
U_3
VCC
D13
Trig
HC-SR04 Ultrasonic Sensor
Echo
D5
GND
R_{15} = 1k
D_{14} = R
R_{16} = 1k
D_{15} = Y
R_{17} = 1k
D_{16} = G
Part of Subsystem 4
D_{17} = G
R_{18} = 1k
D_{18} = R
R_{19} = 1k
5V
U_2
G303BLX Shift Register
1
2
3
4
5
6
7
8
16
15
14
13
12
11
10
9
Transition 1
Transition 3
Transition 2

Sheet 4 - Tunnel height detection
This circuit uses 2 ultrasonic sensors in order to detect overheight vehicles before they enter the tunnel. U1 and U2 cross-check each other at the tunnel entrance (SS4). B18 - Anish Gokulnath Revision 1 7/10/2026 8:23 pm





Sheet 5 - Failure Alert

On the PA2, when Arduino power is lost, an alert is played, using a TL074 compactor which detects the loss and then uses an NE556 timer to generate the tone. The tone is manually switched on using an override switch (OS1).

B18 - Anish Gokulnath
Revision 1
7/10/2026
9:43 pm

